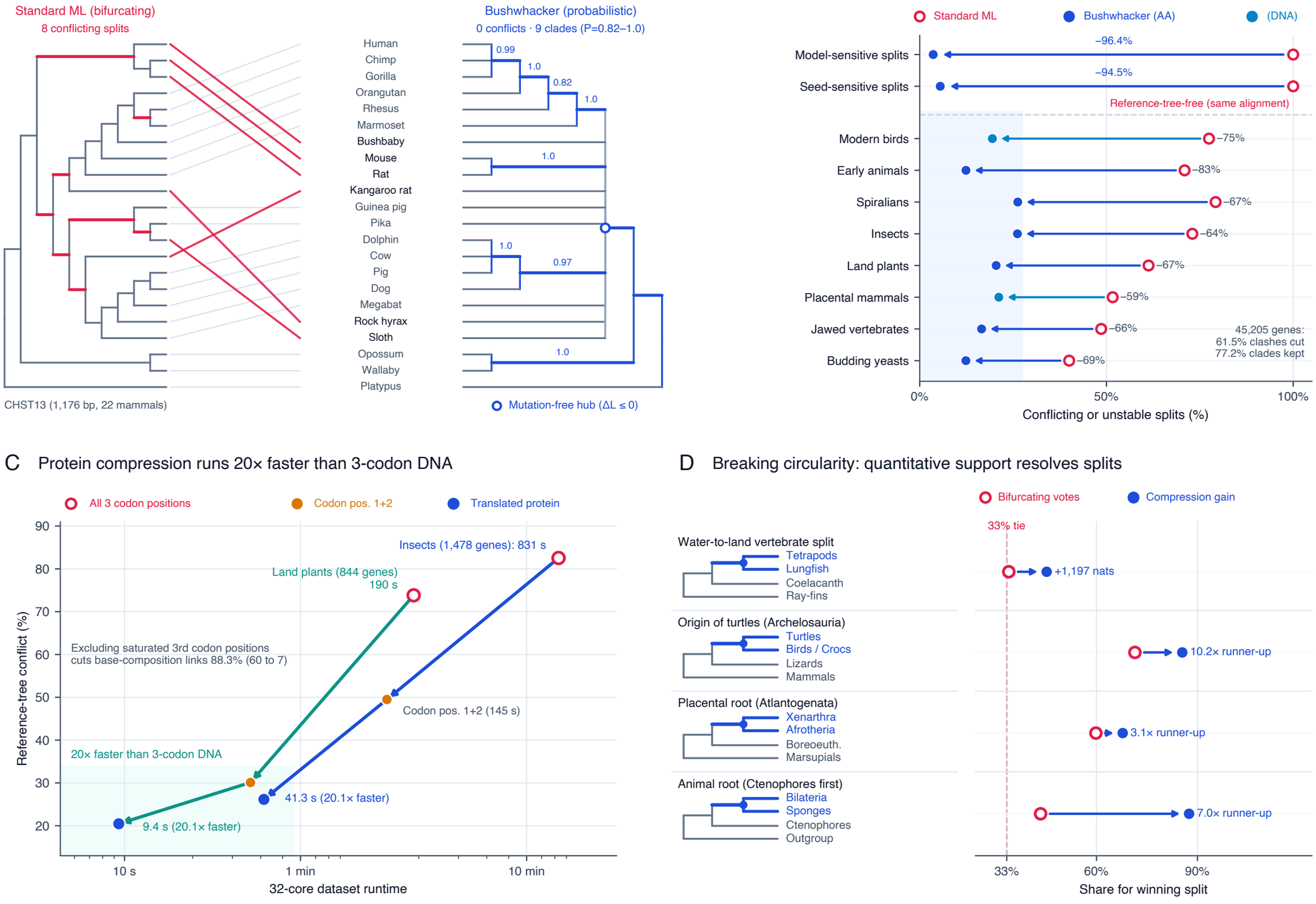
A
Untangling an empirical mammalian gene tree
Standard ML (bifurcating)
8 conflicting splits
Bushwhacker (probabilistic)
0 conflicts · 9 clades (P=0.82–1.0)
Human
Chimp
Gorilla
Orangutan
Rhesus
Marmoset
Bushbaby
Mouse
Rat
Kangaroo rat
Guinea pig
Pika
Dolphin
Cow
Pig
Dog
Megabat
Rock hyrax
Sloth
Opossum
Wallaby
Platypus
0.99
1.0
0.82
1.0
1.0
1.0
0.97
1.0
CHST13 (1,176 bp, 22 mammals)
Mutation-free hub (ΔL ≤ 0)
B
Same-alignment flips and conflict across 19 radiations
Standard ML
Bushwhacker (AA)
(DNA)
Model-sensitive splits
–96.4%
Seed-sensitive splits
–94.5%
Reference-tree-free (same alignment)
Modern birds
–75%
Early animals
–83%
Spiralians
–67%
Insects
–64%
Land plants
–67%
Placental mammals
–59%
Jawed vertebrates
–66%
Budding yeasts
–69%
45,205 genes:
61.5% clashes cut
77.2% clades kept
0%
50%
100%
Conflicting or unstable splits (%)
C
Protein compression runs 20× faster than 3-codon DNA
All 3 codon positions
Codon pos. 1+2
Translated protein
Reference-tree conflict (%)
90
80
70
60
50
40
30
20
Insects (1,478 genes): 831 s
Land plants (844 genes)
190 s
Excluding saturated 3rd codon positions
cuts base-composition links 88.3% (60 to 7)
Codon pos. 1+2 (145 s)
20× faster than 3-codon DNA
41.3 s (20.1× faster)
9.4 s (20.1× faster)
10 s
1 min
10 min
32-core dataset runtime
D
Breaking circularity: quantitative support resolves splits
Bifurcating votes
Compression gain
Water-to-land vertebrate split
Tetrapods
Lungfish
Coelacanth
Ray-fins
Origin of turtles (Archelosauria)
Turtles
Birds / Crocs
Lizards
Mammals
Placental root (Atlantogenata)
Xenarthra
Afrotheria
Boreoeuth.
Marsupials
Animal root (Ctenophores first)
Bilateria
Sponges
Ctenophores
Outgroup
33% tie
+1,197 nats
10.2× runner-up
3.1× runner-up
7.0× runner-up
33%
60%
90%
Share for winning split